

EXPERIMENT 3

EDA-Data Inspection and Analysis

Aim

To inspect and analyze the Customer Churn dataset using DataFrame operations, conditional filtering, descriptive statistics, and measures of central tendency and dispersion.

Procedure

1. Load the Customer Churn dataset using Pandas.
2. View the first and last records of the DataFrame.
3. Inspect the number of rows, columns, and data types.
4. Filter customers based on conditions such as `tenure` and `MonthlyCharges`.
5. Calculate mean, median, and mode.
6. Calculate range, variance, and standard deviation.
7. Display the statistical results.

Code

```
import pandas as pd

# 1. Load dataset
df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv")

# 2. View DataFrame
print("First 5 rows:")
print(df.head())

print("\nLast 5 rows:")
print(df.tail())

# 3. Inspect DataFrame
```

```
print("\nShape:")

print(df.shape)


print("\nColumns:")

print(df.columns)


print("\nData Types:")

print(df.dtypes)


# 4. Filtering data

print("\nCustomers with Monthly Charges > 70:")

print(df[df["MonthlyCharges"] > 70].head())


print("\nCustomers with Tenure > 24 months:")

print(df[df["tenure"] > 24].head())


# 5. Select specific columns

print("\nCustomer ID and Monthly Charges:")

print(df[["customerID", "MonthlyCharges"]].head())


# 6. Mean

mean_value = df["MonthlyCharges"].mean()

print("\nMean:", mean_value)


# 7. Median

median_value = df["MonthlyCharges"].median()

print("Median:", median_value)


# 8. Mode

mode_value = df["MonthlyCharges"].mode()[0]
```

```

print("Mode:", mode_value)

# 9. Range
data_range = (
    df["MonthlyCharges"].max()
    - df["MonthlyCharges"].min()
)

print("Range:", data_range)

# 10. Variance
variance = df["MonthlyCharges"].var()

print("Variance:", variance)

# 11. Standard Deviation
std = df["MonthlyCharges"].std()

print("Standard Deviation:", std)

# 12. Complete descriptive statistics
print("\nDescriptive Statistics:")

print(df["MonthlyCharges"].describe())

```

OUTPUT

First 5 rows:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure
0	7590-VHVEG	Female	0	Yes	No	1
	No					
1	5575-GNVDE	Male	0	No	No	34
	Yes					
2	3668-QPYBK	Male	0	No	No	2
	Yes					
3	7795-CFOCW	Male	0	No	No	45
	No					

4	9237-HQITU	Female	0	No	No	2
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Yes

	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection
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0	No phone service	DSL	No	...	No
1	No	DSL	Yes	...	Yes
2	No	DSL	Yes	...	No
3	No phone service	DSL	Yes	...	Yes
4	No	Fiber optic	No	...	No

	TechSupport	StreamingTV	StreamingMovies	Contract
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PaperlessBilling \

0	No	No	No	Month-to-month
1	No	No	No	One year
2	No	No	No	Month-to-month
3	Yes	No	No	One year
4	No	No	No	Month-to-month

Yes

	PaymentMethod	MonthlyCharges	TotalCharges	Churn
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0	Electronic check	29.85	29.85	No
1	Mailed check	56.95	1889.5	No
2	Mailed check	53.85	108.15	Yes
3	Bank transfer (automatic)	42.30	1840.75	No
4	Electronic check	70.70	151.65	Yes

[5 rows x 21 columns]

Last 5 rows:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	\
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7038	6840-RESVB	Male	0	Yes	Yes	24	
7039	2234-XADUH	Female	0	Yes	Yes	72	
7040	4801-JZAZL	Female	0	Yes	Yes	11	
7041	8361-LTMKD	Male	1	Yes	No	4	
7042	3186-AJIEK	Male	0	No	No	66	

	PhoneService	MultipleLines	InternetService	OnlineSecurity	...
\					
7038	Yes	Yes	DSL	Yes	...
7039	Yes	Yes	Fiber optic	No	...
7040	No	No phone service	DSL	Yes	...
7041	Yes	Yes	Fiber optic	No	...
7042	Yes	No	Fiber optic	Yes	...

	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	
Contract \					
7038	Yes	Yes	Yes	Yes	One
year					
7039	Yes	No	Yes	Yes	One
year					
7040	No	No	No	No	Month-to-
month					
7041	No	No	No	No	Month-to-
month					
7042	Yes	Yes	Yes	Yes	Two
year					

	PaperlessBilling	PaymentMethod	MonthlyCharges
TotalCharges \			
7038	Yes	Mailed check	84.80
1990.5			
7039	Yes	Credit card (automatic)	103.20
7362.9			
7040	Yes	Electronic check	29.60
346.45			
7041	Yes	Mailed check	74.40
306.6			
7042	Yes	Bank transfer (automatic)	105.65
6844.5			

	Churn
7038	No
7039	No
7040	No
7041	Yes
7042	No

[5 rows x 21 columns]

Shape:

(7043, 21)

Columns:

```
Index(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents',  
      'tenure', 'PhoneService', 'MultipleLines', 'InternetService',  
      'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',  
      'TechSupport',  
      'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling',  
      'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn'],  
      dtype='object')
```

Data Types:

customerID	object
gender	object
SeniorCitizen	int64
Partner	object
Dependents	object
tenure	int64
PhoneService	object
MultipleLines	object
InternetService	object
OnlineSecurity	object
OnlineBackup	object
DeviceProtection	object
TechSupport	object
StreamingTV	object
StreamingMovies	object
Contract	object
PaperlessBilling	object
PaymentMethod	object
MonthlyCharges	float64
TotalCharges	object
Churn	object

dtype: object

Customers with Monthly Charges > 70:

```
customerID gender SeniorCitizen Partner Dependents tenure  
PhoneService \
```

4	9237-HQITU	Female	0	No	No	2
Yes						
5	9305-CDSKC	Female	0	No	No	8
Yes						
6	1452-KIOVK	Male	0	No	Yes	22
Yes						
8	7892-POOKP	Female	0	Yes	No	28
Yes						
12	8091-TTVAX	Male	0	Yes	No	58
Yes						

	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	\
4	No	Fiber optic	No	...	No	
5	Yes	Fiber optic	No	...	Yes	
6	Yes	Fiber optic	No	...	No	
8	Yes	Fiber optic	No	...	Yes	
12	Yes	Fiber optic	No	...	Yes	

	TechSupport	StreamingTV	StreamingMovies	Contract
	PaperlessBilling	\		
4	No	No	No	Month-to-month
Yes				
5	No	Yes	Yes	Month-to-month
Yes				
6	No	Yes	No	Month-to-month
Yes				
8	Yes	Yes	Yes	Month-to-month
Yes				
12	No	Yes	Yes	One year
No				

	PaymentMethod	MonthlyCharges	TotalCharges	Churn
4	Electronic check	70.70	151.65	Yes
5	Electronic check	99.65	820.5	Yes
6	Credit card (automatic)	89.10	1949.4	No
8	Electronic check	104.80	3046.05	Yes
12	Credit card (automatic)	100.35	5681.1	No

[5 rows x 21 columns]

Customers with Tenure > 24 months:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure
PhoneService \						
1	5575-GNVDE	Male	0	No	No	34
Yes						
3	7795-CFOCW	Male	0	No	No	45
No						
8	7892-POOKP	Female	0	Yes	No	28
Yes						
9	6388-TABGU	Male	0	No	Yes	62
Yes						
12	8091-TTVAX	Male	0	Yes	No	58
Yes						

	MultipleLines	InternetService	OnlineSecurity	...
DeviceProtection \				
1	No	DSL	Yes	...
Yes				
3	No phone service	DSL	Yes	...
Yes				
8	Yes	Fiber optic	No	...
Yes				
9	No	DSL	Yes	...
No				
12	Yes	Fiber optic	No	...
Yes				

	TechSupport	StreamingTV	StreamingMovies	Contract
PaperlessBilling \				
1	No	No	No	One year
No				
3	Yes	No	No	One year
No				
8	Yes	Yes	Yes	Month-to-month
Yes				
9	No	No	No	One year
No				
12	No	Yes	Yes	One year
No				

	PaymentMethod	MonthlyCharges	TotalCharges	Churn
1	Mailed check	56.95	1889.5	No

3	Bank transfer (automatic)	42.30	1840.75	No
8	Electronic check	104.80	3046.05	Yes
9	Bank transfer (automatic)	56.15	3487.95	No
12	Credit card (automatic)	100.35	5681.1	No

[5 rows x 21 columns]

Customer ID and Monthly Charges:

	customerID	MonthlyCharges
0	7590-VHVEG	29.85
1	5575-GNVDE	56.95
2	3668-QPYBK	53.85
3	7795-CFOCW	42.30
4	9237-HQITU	70.70

Mean: 64.76169246059918

Median: 70.35

Mode: 20.05

Range: 100.5

Variance: 905.4109343405128

Standard Deviation: 30.09004709767854

Descriptive Statistics:

count	7043.000000
mean	64.761692
std	30.090047
min	18.250000
25%	35.500000
50%	70.350000
75%	89.850000
max	118.750000

Name: MonthlyCharges, dtype: float64

Result

The Customer Churn dataset was successfully cleaned by handling missing values, removing duplicates and unnecessary data, converting data types, and applying Min-Max Scaling and Standardization to numerical features.